



APPLIED MACHINE LEARNING FOR TEXT ANALYSIS (24ALT3101)

Machine Learning Framework for Urban Traffic Accident Risk Prediction

Team Members:

Anjali Kasarla - 2420030170
Vinusha Muppala - 2420030212
Lasya Geethika - 2420030678

Under the Esteemed Guidance of:

Dr . KATANGURI SWANTHANA (CSE)



Contents

- Problem Statement
- Objectives
- Literature Survey
- Research Gap
- Innovation & Creativity
- Novelty
- Feasibility
- Project Plan
- Proposed Architecture/Methodology
- Datasets & Technology Stack
- References

PROBLEM STATEMENT

Machine Learning Framework for Urban Traffic Accident Risk Prediction

- Urban traffic accidents are caused by various factors such as **weather, road conditions, traffic density, time, and driver-related causes**. Traditional systems mainly respond after accidents occur and have limited ability to predict accident risk in advance.
- The proposed system uses **historical accident data and machine learning algorithms** such as Logistic Regression, Decision Tree, and Random Forest to identify accident patterns and classify situations as **Low, Medium, or High Risk**. NLP is used to analyze accident descriptions, while **SHAP/LIME** explains the reasons behind predictions.
- The model is evaluated using **Accuracy, Precision, Recall, F1-score, ROC-AUC, and Confusion Matrix** and deployed through a **web dashboard** to support proactive traffic safety decisions.

OBJECTIVES

Objectives :

- To develop a **Machine Learning framework** for predicting urban traffic accident risk.
- To collect and preprocess **traffic accident, weather, road, and traffic-related data**.
- To identify the major **factors contributing to traffic accidents**.
- To apply machine learning algorithms such as **Logistic Regression, Decision Tree, Random Forest, SVM** for risk prediction.
- To classify accident situations into **Low, Medium, and High Risk** categories.
- To evaluate and compare model performance using **Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix**.
- To use **Explainable AI techniques such as SHAP/LIME** to understand the factors influencing predictions.
- To develop a **web-based dashboard** for accident risk prediction and visualization.
- To provide an intelligent system that can support **proactive traffic safety and decision-making**.

LITERATURE SURVEY

No.	Research Paper	Model/Algorithm	Dataset Used	Pros	Cons
1	Citywide Short-Term Crash Risk Prediction – Bao et al. (2019)	STCL-Net, CNN, LSTM, ConvLSTM	NYC crash + taxi + road + weather	Uses multiple data sources ; captures spatial and temporal patterns	Tested mainly in one city ; performance decreases at finer resolutions
2	Real-Time Crash Risk Prediction using LSTM-RNN – Yuan et al. (2019)	LSTM-RNN, SMOTE, Conditional Logistic Model	Traffic detector + crash data	Good for real-time prediction ; handles imbalanced crash data	Requires traffic detector infrastructure ; limited explainability
3	ML and DL Methods for Real-Time Crash Prediction – Theofilatos et al. (2019)	KNN, NB, DT, RF, SVM, NN, DL	Greece traffic + weather + crash data	Compares many ML/DL models	Dataset limited to one tollway ; crash prediction remains difficult
4	Real-Time Crash Prediction in Urban Expressway – Basso et al. (2018)	RF, SVM, Logistic Regression	Chile AVI traffic + crash data	Uses RF, SVM and LR ; suitable for real-time prediction	Requires specialized AVI infrastructure ; high false alarms
5	Highway Crash Detection and Risk Estimation – Huang et al. (2020)	Deep Learning, SVM, Shallow ML	Iowa speed + volume + occupancy + crashes	Uses detailed traffic conditions for risk estimation	Requires roadside radar sensors ; prediction becomes harder farther before crashes

Research Paper

Freeway Crash Risk Prediction using Probe Vehicle Data – Zhang et al. (2022)

Motorway Segment Crash Risk Level Prediction – Nikolaou et al. (2023)

Traffic Accident Severity Prediction Based on Random Forest – Yan & Shen (2022)

Enhanced MSCPO-XGBoost Accident Severity Prediction – Chen et al. (2025)

Calibrated & Explainable Gradient Boosting – Alhawarat et al. (2026)

Model / Algorithm

RF, SVM, XGBoost

LR, DT, RF, SVM, KNN, SHAP

RF, Bayesian Optimization, PDP

XGBoost, SVM, RF, CNN, SHAP, SMOTE

XGBoost, LightGBM, SHAP

Dataset Used

HERE probe-vehicle + crash data

Greece motorway data

US-Accidents

China NAIS, 2018–2023

US-Accidents + UK STATS19

Pros

Uses **crowdsourced vehicle data**; useful for real-time prediction

Risk-level prediction + SHAP explainability; compares models

Large and widely used accident dataset; strong **RF classification**

Combines **ML + SHAP + SMOTE**; handles imbalance

Large datasets + explainability + calibration; strong recent approach

Cons

Probe vehicles may not represent all vehicles; limited traffic variables

Limited to **one motorway**; limited real-time weather/traffic information

Predicts **severity rather than future risk**; limited geographical scope

Relatively small dataset; China-specific; focuses mainly on severity

Severity definition differs between datasets; transfer to new regions may be difficult

Real-World Applications & Existing Systems

Existing System

Google Maps

Waze

Road Accident Prediction Models

Accident Severity Prediction Systems

CCTV-Based Accident Detection

Description

Provides navigation, traffic updates, and estimated arrival time.

Displays accidents and incidents reported by users.

Predict accident occurrence using historical datasets.

Predict how severe an accident will be.

Detects accidents using cameras and deep learning.

Limitations

Focuses on fastest route, not accident risk. Does not explain safety level.

Depends on user reports. Cannot predict future accident risk.

Mostly rely on historical data; limited contextual information.

Useful mainly after an accident has occurred.

Requires expensive camera infrastructure and detects accidents after they occur.

RESEARCH GAP

1. Fragmented Accident Prediction Approaches

Existing studies often analyze **weather, road conditions, traffic, and accident factors separately**, rather than combining them into one unified prediction framework.

2. Limited Use of Textual Accident Information

Most accident prediction models primarily use **structured/tabular data**, with limited use of accident descriptions and other textual information for risk prediction.

3. Limited Explainability

Many machine learning models provide an accident-risk prediction but do not clearly explain **which factors contributed to the prediction**.

4. Lack of Integrated Risk Classification

Existing approaches often focus on **accident occurrence or severity**, while fewer systems provide an integrated **Low, Medium, and High risk classification** for proactive traffic safety.

5. Limited Real-Time Decision Support

Many models are evaluated offline and are not integrated into an accessible **web-based dashboard** for practical traffic-risk monitoring and decision-making.

INNOVATION & CREATIVITY

01 — Unified Accident Risk Framework

Combines **traffic, weather, road, and accident-related factors** into a single prediction system.

Provides an integrated approach to urban accident-risk assessment.

02 — NLP-Based Accident Analysis

Uses **text preprocessing and feature extraction** to analyze accident-related textual information.

Goes beyond simple structured-data analysis.

03 — Explainable Risk Prediction

Uses **SHAP/LIME** to identify the factors that influence accident-risk predictions.

Makes the ML model more **transparent and understandable**.

04 — Interactive Risk Dashboard

Provides **Low, Medium, and High risk predictions** through a web-based interface.

Supports traffic monitoring, visualization, and **proactive decision-making**.

Key Idea:

AI does not only predict traffic accident risk — it explains the reasons behind the prediction and supports safer traffic management.

NOVALITY



Integrated Risk Prediction

Combines weather, traffic, road, and accident factors to predict urban accident risk in a unified framework.



NLP-Enhanced Accident Analysis

Uses text preprocessing and feature extraction to extract useful information from accident-related textual data.

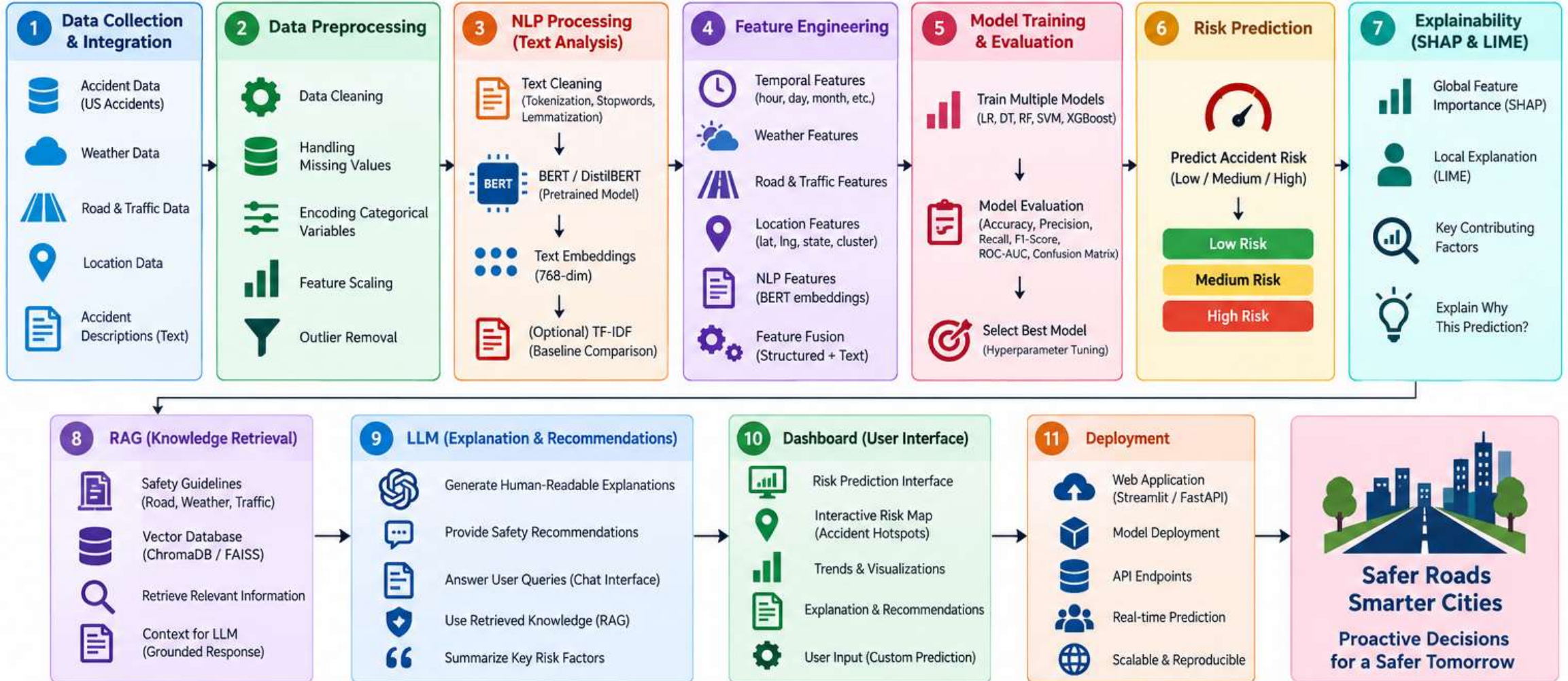


Explainable AI-Based Prediction

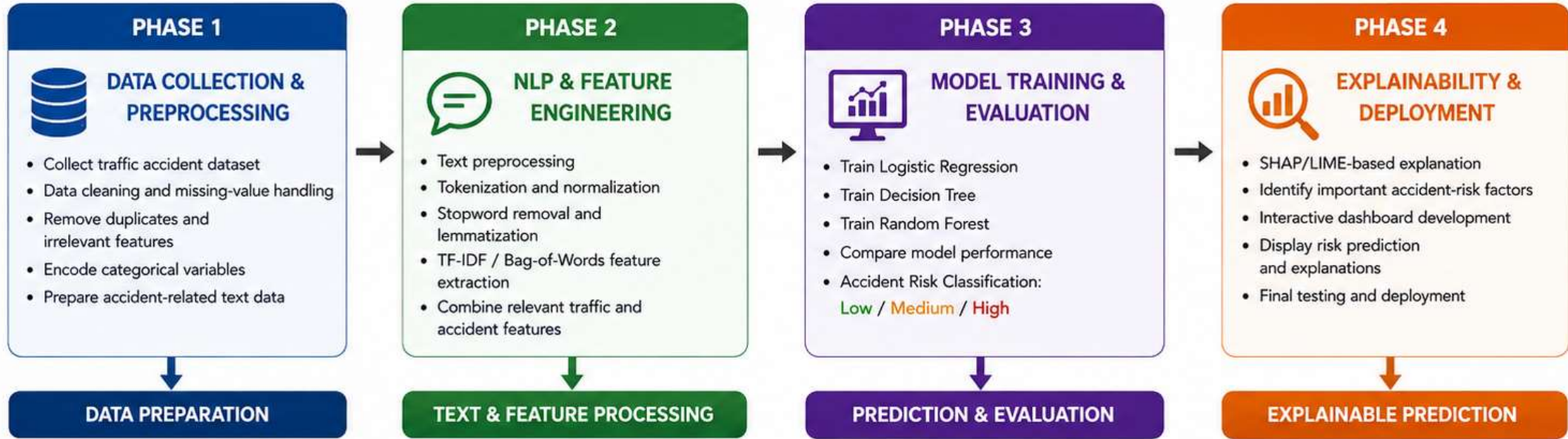
Uses SHAP/LIME to identify the key factors responsible for each accident-risk prediction.

FEASIBILITY ANALYSIS

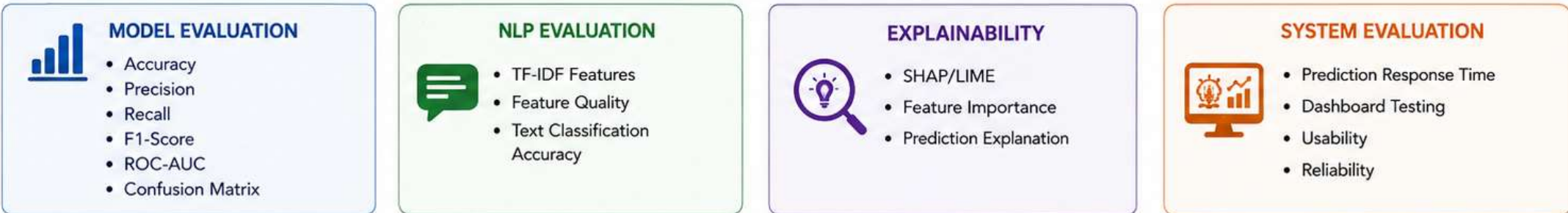
From Data to Decisions: Predict • Explain • Assist • Prevent



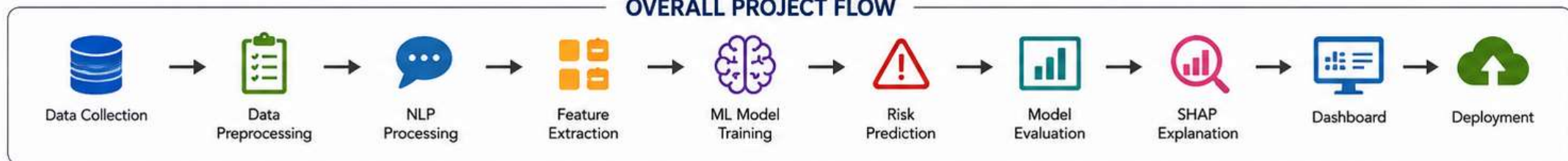
PROJECT PLAN



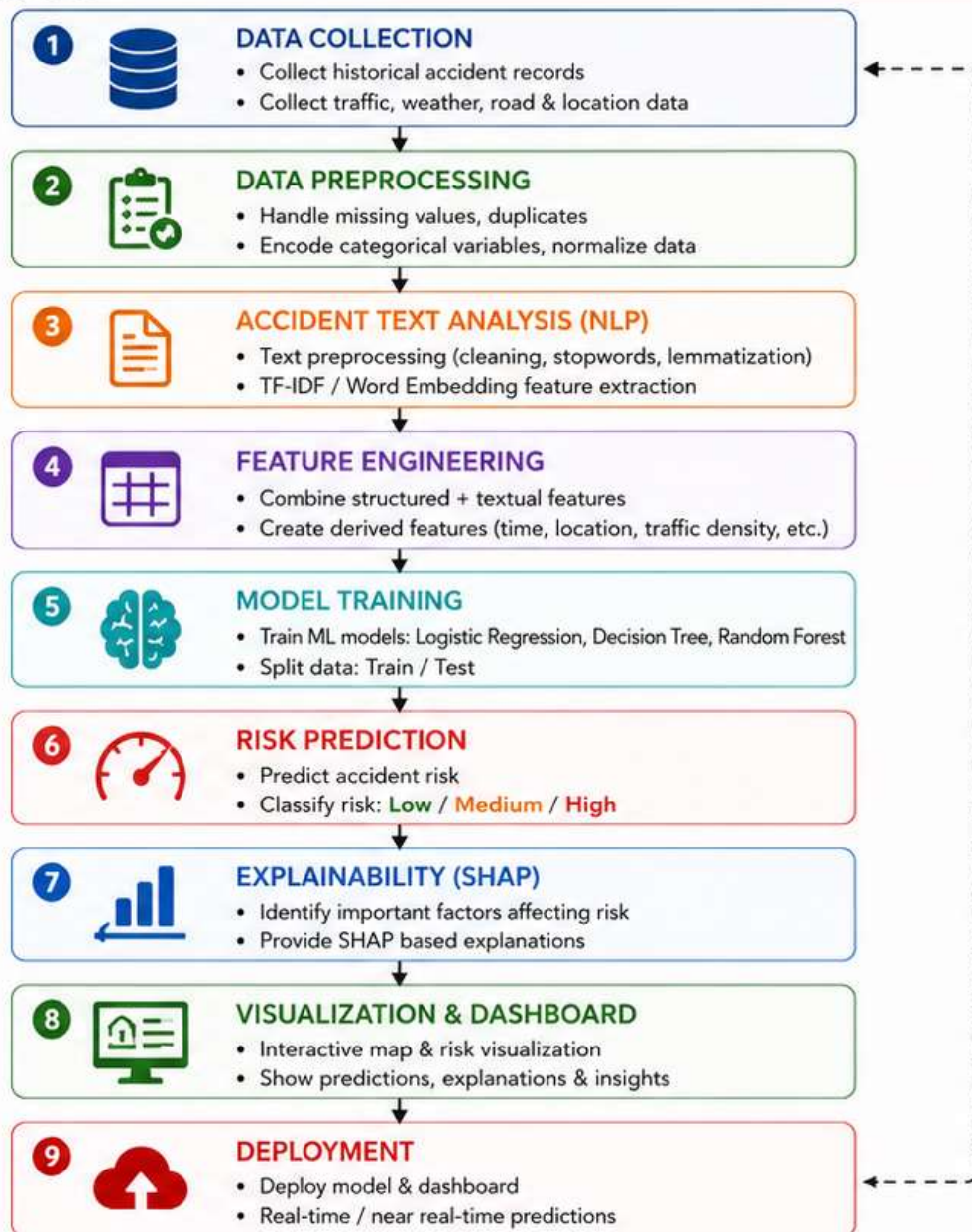
EVALUATION METRICS



OVERALL PROJECT FLOW



PROPOSED ARCHITECTURE/METHODOLOGY



Workflow:

- 1 Data Collection**
Collect historical accident data along with traffic, weather, road and location information.
- 2 Data Preprocessing**
Clean and prepare data by handling missing values, encoding categorical variables and normalizing features.
- 3 Accident Text Analysis (NLP)**
Process accident descriptions using NLP techniques and extract meaningful text features.
- 4 Feature Engineering**
Combine structured and textual features and create new informative features for better prediction.
- 5 Model Training**
Train machine learning models and choose the best performing model.
- 6 Risk Prediction**
Predict urban traffic accident risk and classify it into **Low**, **Medium** or **High**.
- 7 Explainability (SHAP)**
Use SHAP to identify key factors responsible for each prediction.
- 8 Visualization & Dashboard**
Display predictions, risk levels, important factors and trends on an interactive dashboard.
- 9 Deployment**
Deploy the model and dashboard for real-time risk prediction and decision support.



Thankyou